



沧州宏宇石油机械有限责任公司
CANGZHOU HONGYU PETROLEUM MECHINERY CO., LTD

TCNJ-系列泥浆搅拌机 使用说明书

沧州宏宇石油机械有限责任公司
CANGZHOU HONGYU PETROLEUM MECHINERY CO., LTD
2024



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如果您在使用中有什么疑问，或遇到了什么问题
请及时与我们联系，我们将热情为您服务。

If you have any questions or problems, please contact us, we shall serve
for you enthusiastically!

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一、用途及适用范围

钻井液搅拌器是钻井液固控系统的重要组成部分。它安装在钻井液循环罐面上，叶轮插入钻井液面下一定的深度，直接搅动钻井液。其作用是：（1）、通过叶轮的搅拌使加入到钻井液里的化学药品、粘土等均匀混合，并使化学药品等有用固相快速分散，提高钻井液的粘度和静切力；（2）通过搅拌器的搅动防止钻井液固相发生沉淀，使钻井液保持稳定的性能，保证钻井工程的安全顺利进行。

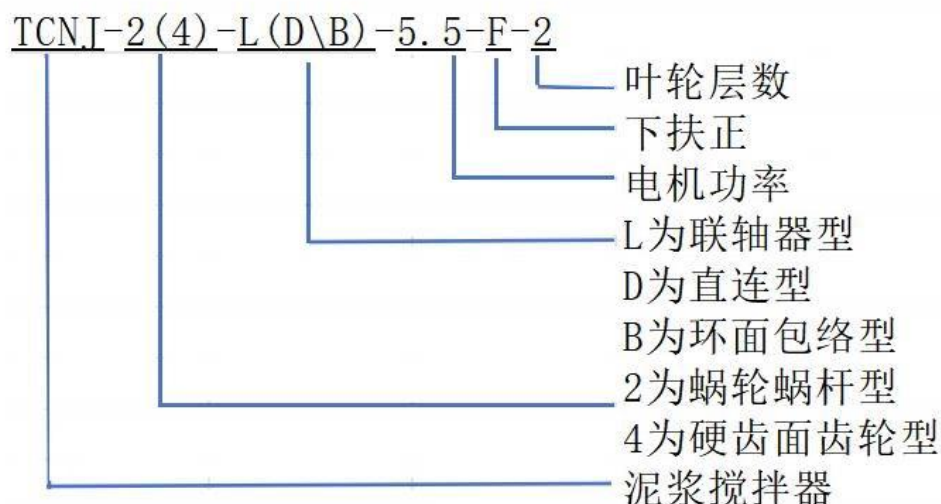
二、结构及工作原理

TCNJ 系列钻井液搅拌器主要由防爆电机、减速器、底座、护罩、联轴器、叶轮轴、叶轮等部件组成。其工作原理是：电机驱动减速器的输入轴，减速后通过联轴器带动叶轮轴转动，钻井液在叶轮的搅拌下，形成循环流，使整个罐里的钻井液性能保持稳定。

三、特点

该搅拌器电机采用卧式安装，减速机构采用单级圆弧圆柱蜗轮蜗杆减速机构，具有高效、耐用的特点；输入输出轴头采用了进口油封，使轴封更可靠、工作寿命更长；该搅拌器还具有重心低、稳定性好、运转平稳、搅拌效果好、外型美观等优点。

四、代号示例



五、主要技术参数（标准参数）

产品型号	额定功率	叶轮转速	叶片数量	叶轮直径	叶轮倾角	搅拌长度	推荐搅拌容积	
TCNJ	4.0kw	72r/min	4/8	920mm	45°	800-1800mm	40m³ 罐安 3台, 60m³ 罐安装 5 台	
	<u>5.5kw</u>					200-2200mm		
	7.5kw							
	11kw			上:				
	15kw			920mm				
	<u>18.5kw</u>			下:				
	22kw			430mm				

六、安装（仅供参考）

1、安装尺寸

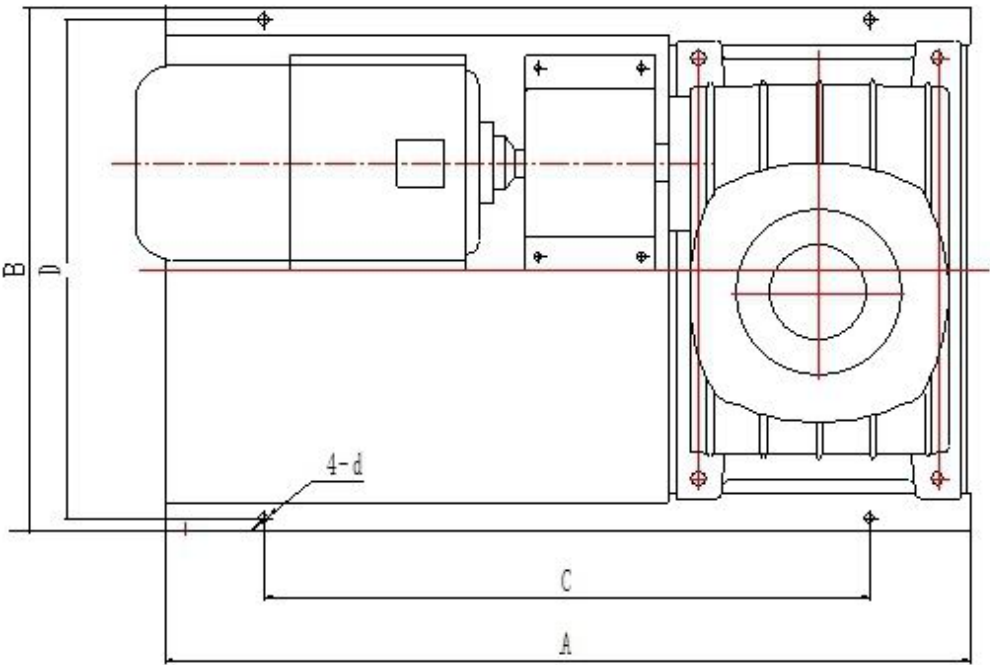


图1

安装尺寸 型号					
	A	B	C	D	4-Md
TCNJ-4.0	870	521	765	433	4-M12*40
TCNJ-5.5	1040	570	905	468	4-M16*40
TCNJ-7.5	1080	570	905	468	4-M16*40
TCNJ-11	1240	650	1090	564	4-M16*40
TCNJ-15	1310	650	1090	590	4-M16*40
TCNJ-18.5	1505	880	1270	774	4-M20*40
TCNJ-22	1540	880	1270	774	4-M20*40



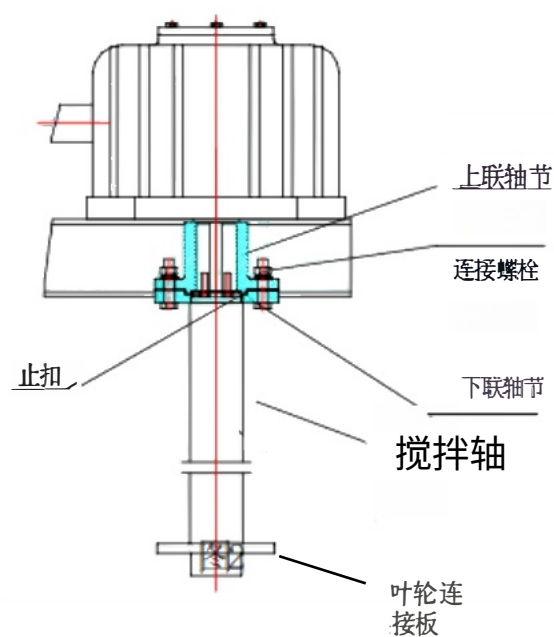
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2、安装程序：

(1) 将搅拌器底座（包括电机及减速机）摆放到罐面搅拌器安装位置，如图 1 所示，将 4 个 M20 的安装块与罐面焊接，用 M20*40 的螺丝连接牢固；搅拌器尽量调整水平。

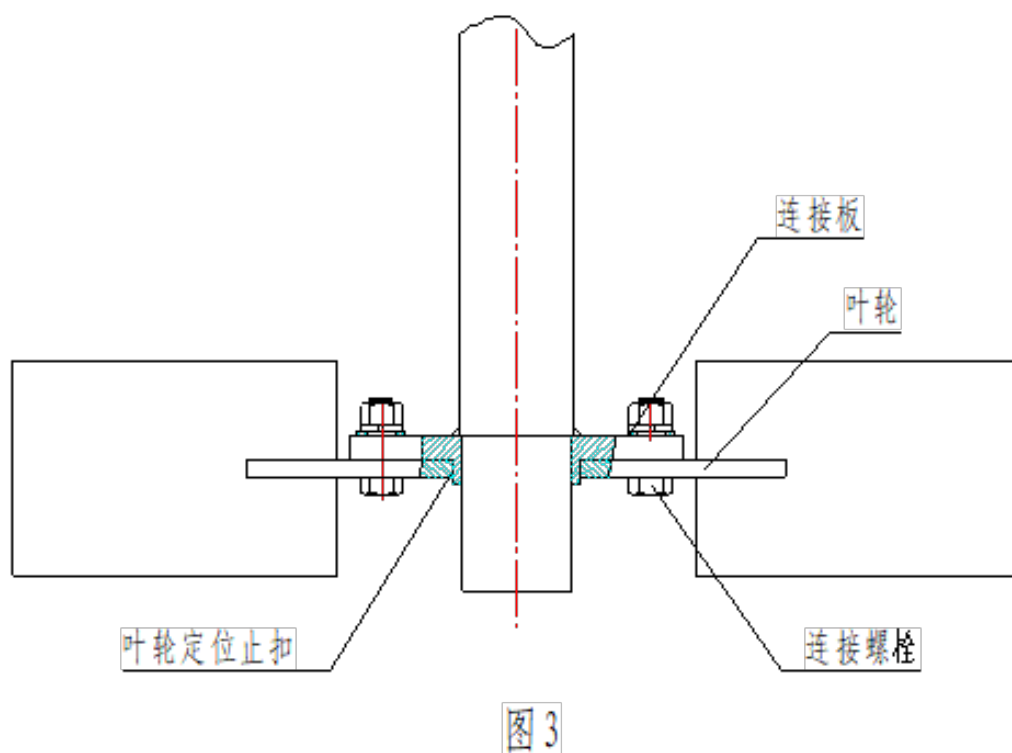
(2) 将搅拌杆与减速机输出轴法兰连接，如图 2 所示，通过 8 条 M12 的螺栓将搅拌器上的法兰与减速机输出轴上的法兰连接到一起，特别注意一定将止口对好，以免转轴偏斜；





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(3) 安装叶轮，如图 3 所示，通过 8 条 M12 的螺栓将搅拌杆上的连接板与叶轮内孔止口对齐，连接到一起，特别注意一定将止口对好，以免叶轮与搅拌杆不同心。





七、调试及维护

1、搅拌器安装好后，在运转以前一定要先做好以下两项工作：（1）.加注 N220~N320（环境温度-30~40℃）或 N320~N460（环境温度 40~65℃）蜗轮蜗杆润滑油至油标中心点（加注量按照提示，用油尺检测）；（2）.取掉通气器上的红色小锥塞（保持内外气压）；

2、点按启动器，观察减速机输入轴的转向是否与箭头所标的方向一致，如果不一致，通过调整电机接线盒上任意 2 个接线柱；

3、新搅拌器第一次加油后，运转 100 小时，使蜗轮蜗杆充分磨合后，在停机的状态下打开减速机侧面的放油塞，放净里面的机油，再换上新油。以后每 2500 小时换油一次（用油尺检查油的颜色及粘度，发生变黑、水化及时更换）；

4、定期检查减速器的油面高度，发现过低时，应及时补充至油标中心点；

5、定期紧固搅拌器的所有连接螺栓，以防松动后造成部件损坏。

八、操作：

1、新安装的搅拌器首先要确认转向，点按启动器，观察减速机输入轴的转向是否与箭头所标的方向一致，如果不一致，通过调整电机上任意 2 个接线柱；

2、确认转向正确后，即可启动，使搅拌器进入正常工作；

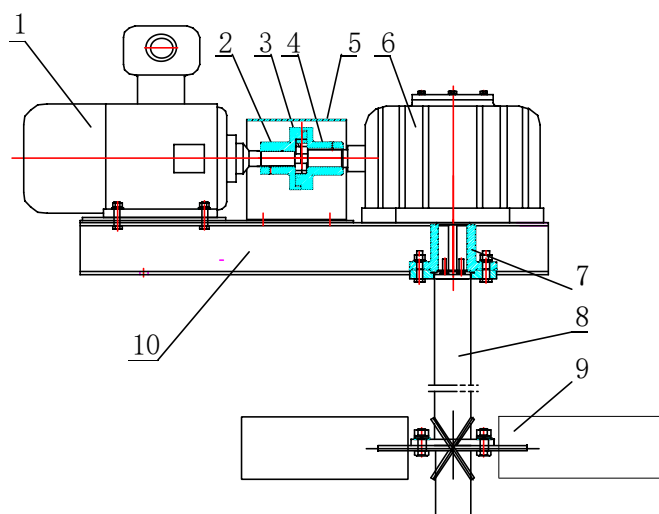
3、如果发现搅拌叶轮上有缠绕物，一定要先停机再做处理。

4、当泥浆中有固相沉淀严重时，启动前要先用泥浆枪冲刷稀释叶轮周围的沉淀物，然后再启动电机。否则会造成严重后果。

九、主要零部件及易损件



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1. 电机; 2. 电机联轴节; 3. 弹性胶块; 4. 减速机联轴节; 5. 护罩; 6. 减速机; 7. 上联轴节; 8. 转轴总成; 9. 叶轮; 10. 底座

序号	零部件名称	图号							数量
		TCNJ-4.0	TCNJ-5.5	TCNJ-7.5	TCNJ-11	TCNJ-15	TCNJ18.5	TCNJ22	
1	电机	4.0	5.5	7.5	11	15	18.5	22	1
2	电机联轴节	4.0-01	5.5-01	5.5-01	11-01	15-01	18.5-01	22-01	1
3	联轴器胶块	4.0-02	5.5-02	5.5-02	11-02	15-02	18.5-02	22-02	1
4	减速机联轴节	4.0-03	5.5-03	5.5-03	5.5-03	15-03	18.5-03	22-03	1
5	护罩	4.0-04	5.5-04	5.5-04	11-04	15-04	18.5-04	22-04	1
6	减速机	135	150	150	175	200	250	250	1
7	上联轴节	4.0-05	5.5-05	5.5-05	11-05	15-05	18.5-05	22-05	1
8	搅拌轴	4.0-06	5.5-06	5.5-06	11-06	15-06	18.5-06	22-06	1
9.1	大叶轮	4.0-07	5.5-07	5.5-07	11-07	15-07	18.5-07	22-07	1
9.2	小叶轮			5.5-08	11-08	15-08	18.5-08	22-07	1
10	底座	4.0-09	5.5-09	5.5-09	11-09	15-09	18.5-09	22-09	1
11	输入轴油封	40*65*12	50*75*12		50*75*12	65*90*12			1
12	输出轴油封	60*85*12	65*90*12		70*95*12	75*105*12			2
13	油镜	M27	M27		M27	M27	M27		1
14	透气帽	M20	M20		M20	M20	M20		1
15	黄油嘴	M10	M10		M10	M10	M10		1
16	入轴轴承		7212		7514	7313			2
17	出轴轴承		7213		7214	7215			2
			6213		6214	6215			1

注：用户可根据实际需要，向生产厂订购以上易损件。



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十、常见故障及排除方法

故障现象	原 因	排除方法
轴转不动	1、电机缺相 2、减速箱被卡死	1、检查电机线路 2、检修减速箱
减速箱响声大	1、缺润滑油 2、减速箱中进入磨粒性杂质或者蜗轮蜗杆啮合有问题 3、减震胶垫损坏 4、输入轴轴承损坏	1、检查油位并添加润滑油 2、检修减速箱，更换润滑油 3、更换减震垫 4、更换轴承
搅拌区域内有沉淀	搅拌杆转向反	调节电机转向与护罩上的箭头方向一致
振动严重	1、叶轮转轴偏心过大 2、搅拌器底座固定不牢固 3、电机轴与减速机轴不同心 4、搅拌器整体不水平 5、底部扶正器不同心	1、校正叶轮转轴 2、上紧固定螺栓 3、调节同轴度 4、调整底座 5、调整扶正器，或直接去掉
漏油	1、输入轴漏油 2、输出轴漏油 3、透气孔漏油 4、放油塞漏油 5、箱体漏油	1、更换油封 2、降低油面 3、拧紧放油塞，螺纹加密封胶 4、更换箱体或用铸工胶修补



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TCNJ Mud Agitator Operation Manual

CANGZHOU HONGYU PETROLEUM MECHINERY CO. , LTD

(PLEASE READ THE MANUAL CAREFULLY BEFORE USE)

2024



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1. Usage and Applicability

Mud agitator is an important part of the drilling-fluid solids controlling system. It can be installed on drilling fluid tank, with the impeller submerged into certain depth under fluid surface to stir the fluid directly. During this process, the drilling fluid can be mixed even, and the solid particles are eliminated. In this way, it can improve solid phase dispersion, and increase the viscosity and gel strength, thus to make drilling fluid keep accordant with the requirement, provide needful fluid for drilling process, and ensure the smooth going of drilling work.

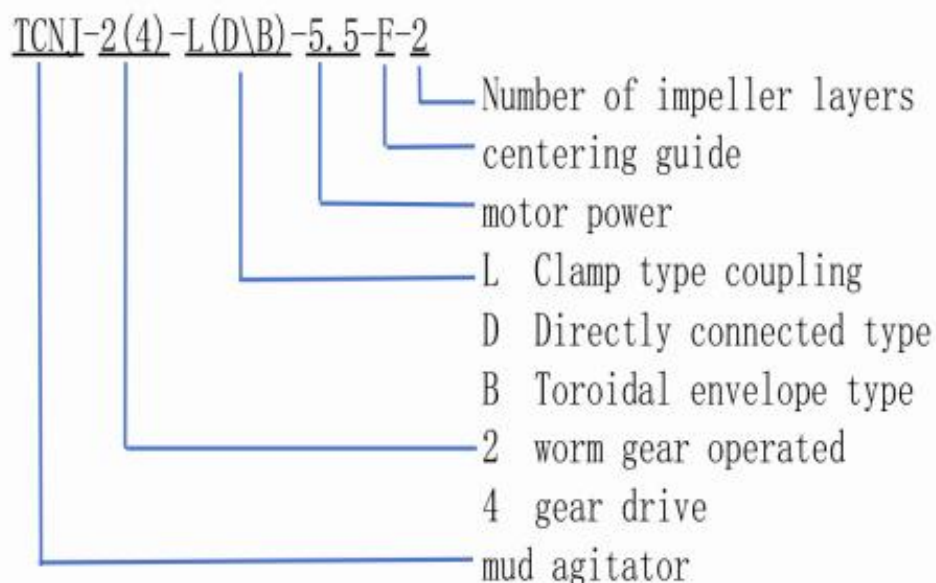
2. Working Mechanism

Driven by the motor and after slowed down by the deceleration, agitator shaft drives impeller to rotate, then drilling fluid forms cycle flow under certain rotary speed of impeller, and makes the solid particle of drilling fluid suspend. Drilling fluid density and its properties can keep steady under successive stirring.

3. Characteristic

The agitator will be installed horizontally. Its deceleration setup, simplex turbine are of high efficiency, and there is special setup to ensure bearing seal reliable. Furthermore, the agitator is of good stability and shock attenuation, and its small appearance can save room for tank surface. In brief, it can operate with good effect and high efficiency.

4、Code name:



5. Main parameters:

Parameters Model	Power kW	Impeller rotate speed /rpm		Impeller diameter/ mm	
		50Hz	60Hz	上	下
TCNJ-4.0	4.0	72	84	750	
TCNJ-5.5	5.5	72	84	800	
TCNJ-7.5	7.5	72	84	920	460
TCNJ-11	11	72	84	920	460
TCNJ-15	15	72	84	920	460
TCNJ-18.5	18.5	72	84	960	640
TCNJ-22	22	72	84	960	640

TCNJ mud agitator can be applied to solid phase controlling of various drilling fluid, and can match with different shale shakers, desanders and desilters.

6. Installation （ For reference only ）

a) Installation dimensions

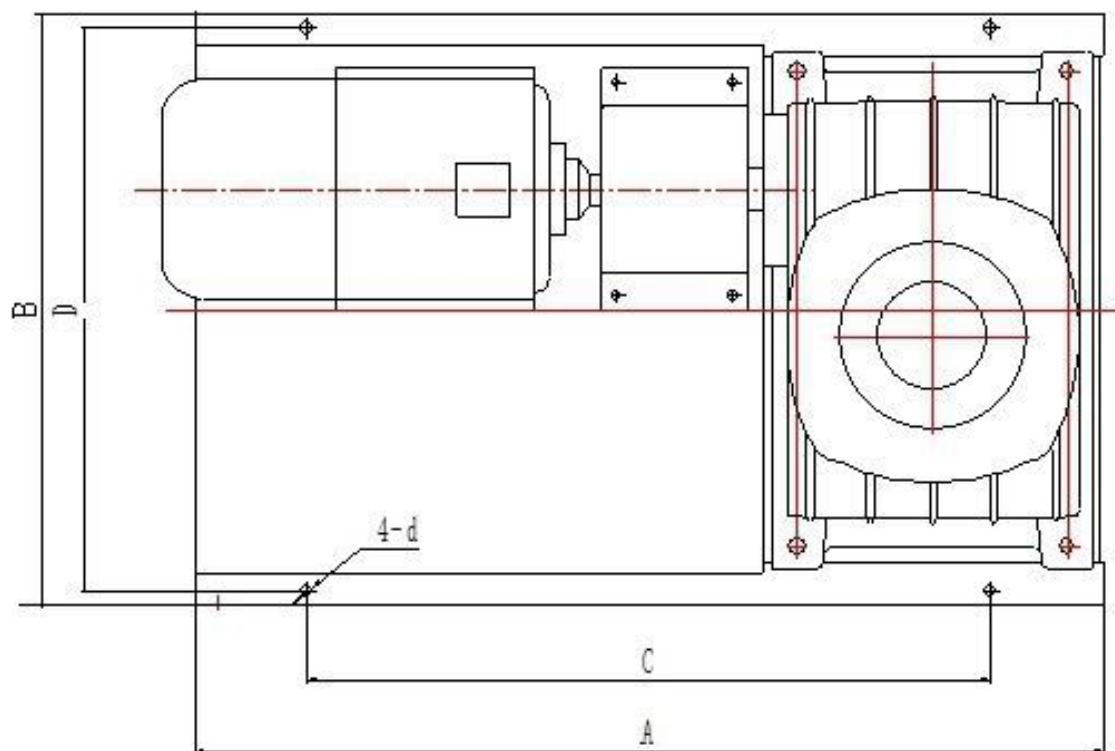


图1



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Installation dimensions Model	A	B	C	D	4-Md
TCNJ-4.0	870	521	765	433	4-M12*40
TCNJ-5.5	1040	570	905	468	4-M16*40
TCNJ-7.5	1080	570	905	468	4-M16*40
TCNJ-11	1240	650	1090	564	4-M16*40
TCNJ-15	1310	650	1090	590	4-M16*40
TCNJ-18.5	1505	880	1270	774	4-M20*40
TCNJ-22	1540	880	1270	774	4-M20*40

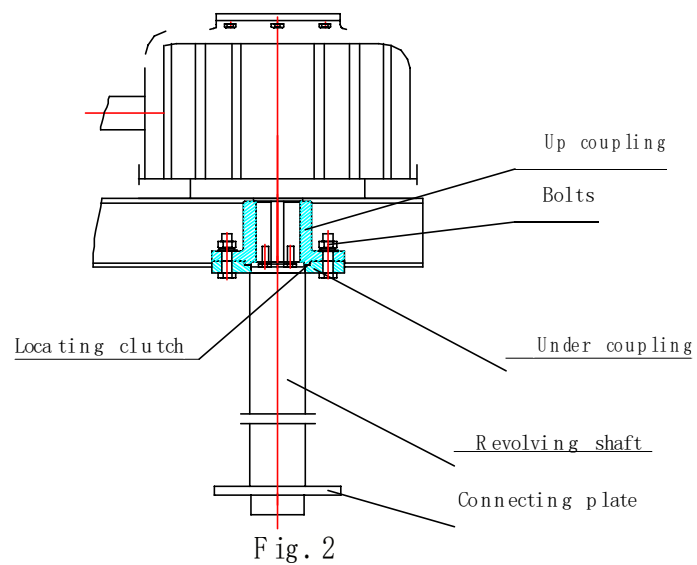
b) Installation procedure:

(1) Place the agitator base (including motor and reducer) on the installation position of the agitator on the tank surface, as shown in Figure 1. Weld four M20 mounting blocks to the tank surface, and connect them firmly with M20*40 screws; Adjust the blender level as much as possible.

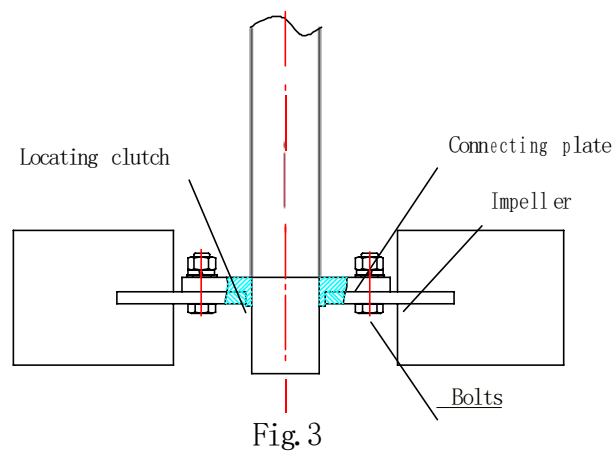
(2) Connect the revolving shaft with the reducer.As shown in Fig.2,connect the under coupling with the up coupling by M12 bolts. Be sure the locating clutch is in alignment.



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(3) Install the impeller, as shown in Fig.3, connect the connecting plate of revolving shaft with the flange of impeller. Be sure the locating clutch is in alignment.





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7. Commissioning and Maintenance

(1) After installing the agitator, be sure that adding lubricant oil N220~N320 or N320~N460 up to the oil pointer centre point, and taking off the red tapered plug on the breather.

(2) Click the starter and watch the rotate direction. If the rotate direction is not in line with the arrow, adjust any 2 contractpr post of the motor.

(3) After adding lubricant oil, run 100 hours, and then stop to let out oil, and add new lubricant oil. Be sure that chang oil every 2500 hours.

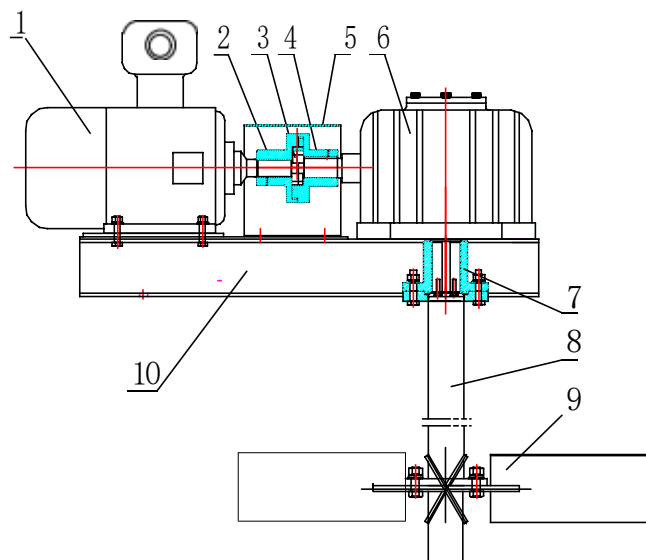
(4) Check the oil head at regular intervals, be sure the oil head is at the oil pointer centre point.

(5) Check the connecting bolts at regular intervals, be sure all the bolts are tighten.

8. Operation

- a) Before working making agitator fan rotation director correct.
- b) After starting agitator judge working noise.
- c) If discover something tangle fans, stop agitator and deal with it.

9. Main parts & wearing parts



motor 2. motor coupling 3. rubber block 4. reducer coupling 5. shield
reducer 7. upper coupling 8. revolving shaft assembly 9. impeller 10. base



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Item	Name	Drawing No.							Quantity
		TCNJ-4.0	TCNJ-5.5	TCNJ-7.5	TCNJ-11	TCNJ-15	TCNJ18.5	TCNJ22	
1	Motor	4.0	5.5	7.5	11	15	18.5	22	1
2	Motor coupling	4.0-01	5.5-01	5.5-01	11-01	15-01	18.5-01	22-01	1
3	Rubber for Coupling	4.0-02	5.5-02	5.5-02	11-02	15-02	18.5-02	22-02	1
4	Reducer coupling	4.0-03	5.5-03	5.5-03	5.5-03	15-03	18.5-03	22-03	1
5	Shield	4.0-04	5.5-04	5.5-04	11-04	15-04	18.5-04	22-04	1
6	reduction box	135	150	150	175	200	250	250	1
7	Upper coupling	4.0-05	5.5-05	5.5-05	11-05	15-05	18.5-05	22-05	1
8	stirring shaft	4.0-06	5.5-06	5.5-06	11-06	15-06	18.5-06	22-06	1
9	big impeller	4.0-07	5.5-07	5.5-07	11-07	15-07	18.5-07	22-07	1
10	Small impeller			5.5-08	11-08	15-08	18.5-08	22-07	1
11	pedestal	4.0-09	5.5-09	5.5-09	11-09	15-09	18.5-09	22-09	1
12	Input shaft oil seal	40*65*12	50*75*12		50*75*12	65*90*12			1
13	Output shaft oil seal	60*85*12	65*90*12		70*95*12	75*105*12			2
14	Oil immersion lens	M27	M27		M27	M27	M27		1
15	Bleeder cap	M20	M20		M20	M20	M20		1
16	grease fitting	M10	M10		M10	M10	M10		1
17	Entry bearing		7212		7514	7313			2
18	Output shaft bearing		7213		7214	7215			2
			6213		6214	6215			1



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10. Trouble removal

Fault phenomenon	Reason	Elimination methods
Non-rotation of shaft	1、MOTOR PHASE 2、The reducer is stuck	1、Check motor circuit 2、Repair gear box
The gearbox is making a loud noise	1、Lack of lubricating oil 2、There are impurities in the gearbox or worm gear meshing problems 3、The shock absorbing rubber pad is damaged 4、The input shaft bearing is damaged	1、Check the oil level and add lubricating oil 2、Check the gearbox and replace the lubricating oil 3、Replace the shock pad 4、Change bearing
There is precipitation in the mixing area	Mixing rod turned backwards	Adjust motor steering in the same direction as the arrow on the shield
Severe vibration	1、Impeller shaft eccentricity is too large 2、Agitator base is not securely fixed 3、The motor shaft is different from the reducer shaft 4、The mixer is not level overall 5、Bottom centralizer different center	1、Calibrate the impeller shaft 2、Tighten the retaining bolt 3、Adjusting coaxiality 4、Adjusting base 5、Adjust the centralize, Or just remove
Oil leak	1、Input shaft leakage 2、Output shaft leakage 3、Vent leakage 4、Oil drain plug leaking 5、Tank leak	1、Replace oil seal 2、Lower oil level 3、Tighten the drain plug and add sealant to the thread 4、Replace case or repair with casting glue